

Lecture 7-1: Introduction to quantum mechanics-3

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Classroom: Teaching Center 401

Wednesday/Friday: 10:15-11:00; 11:10-11:55

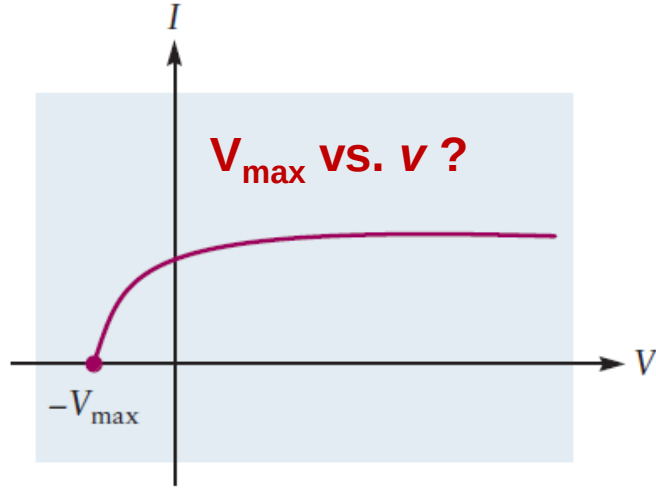
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Major points of last lecture

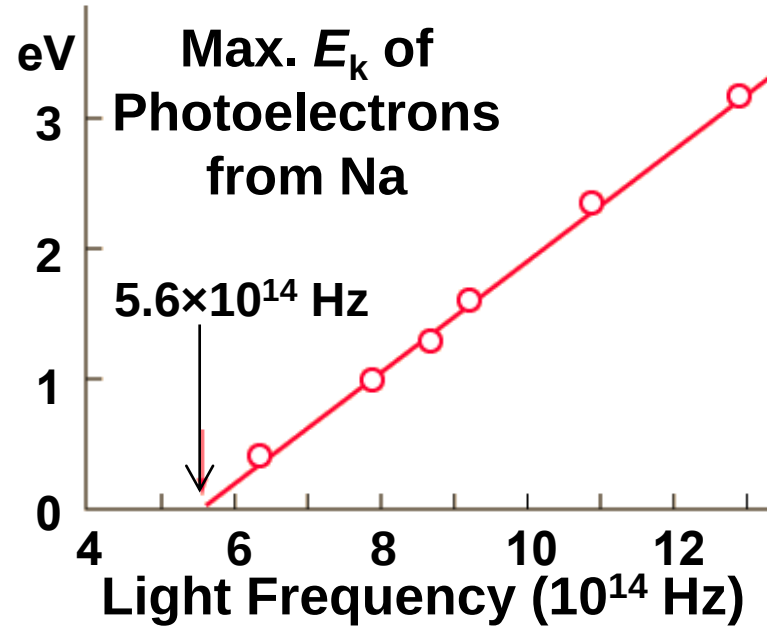
- **Introduction to quantum mechanics**

1. Preliminaries: wave motion and light
2. Evidence for energy quantization in atoms
3. The Bohr model: predicting discrete energy levels in atoms
4. Evidence for wave-particle duality
5. The Schrödinger equation
6. Quantum mechanics of particle-in-a-box models
7. Wave functions for particles in two- and three-dimensional boxes

The Photoelectric Effect



The current in a photocell depends on the potential between cathode and collector.



Albert Einstein
(1879–1955)

1st: a light wave of frequency ν consists of quanta of energy (later called **photons** 光子 by G. N. Lewis), each of which carries energy,
 $E_{\text{photon}} = h\nu$.

2nd: An electron in the metal absorbs a photon of light and thereby gains the energy required to escape from the metal.

1905: Albert Einstein's Year of Miracles

$$E_{\text{light}} = h\nu$$

$$E_{k,\max} + \phi = E_{\text{light}} \propto \nu$$

wave-particle duality of light

(1) Planck 1900, Einstein 1905:

$$E = h\nu$$

(2) Einstein 1916:

$$E = mc^2$$

(3) Definition of momentum

$$p = mv, \quad v = c \text{ for light}$$

(1)+(2)+(3)

$$p = \frac{h\nu}{c} = \frac{h}{\lambda}$$

Photo- + -on = Photon 光子

Standing Waves

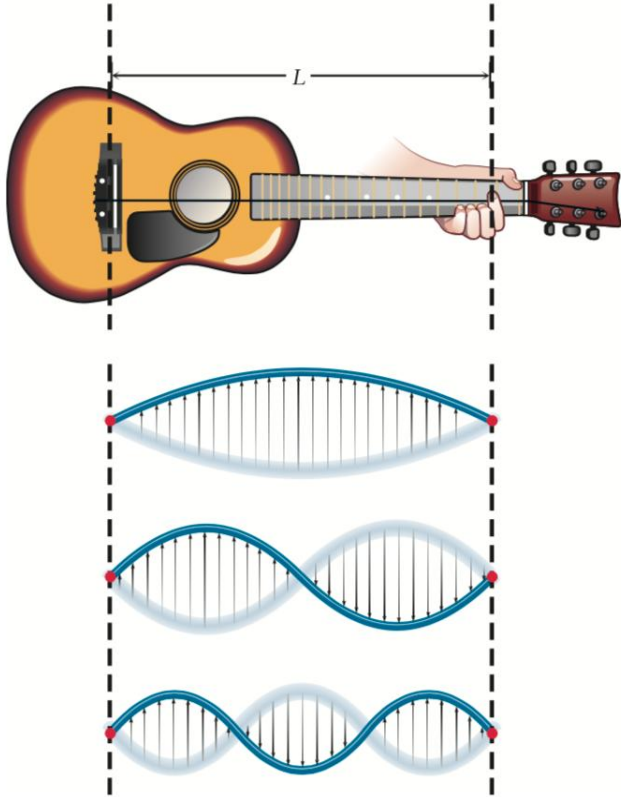


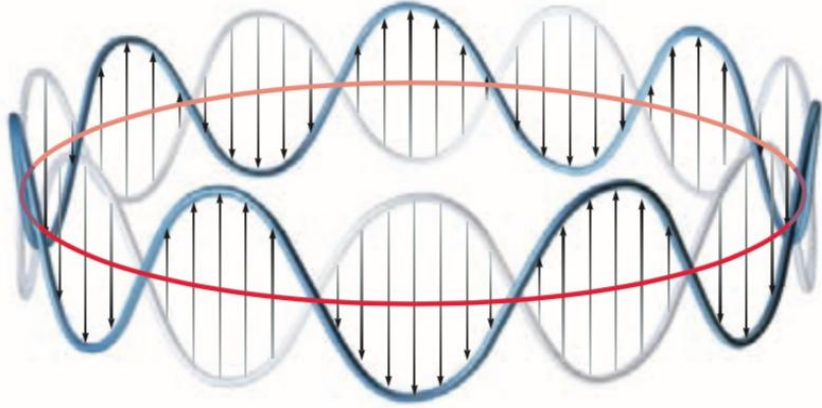
FIGURE 4.17 A guitar string of length L with fixed ends can vibrate in only a restricted set of ways. The positions of largest amplitude for the first three harmonics are shown here. In standing waves such as these, the whole string is in motion except at its end and at the nodes.

The only allowed condition is: an integral number of half-wave-lengths fits into the length of string, L .

$$n \frac{\lambda}{2} = L \quad n = 1, 2, 3, \dots$$

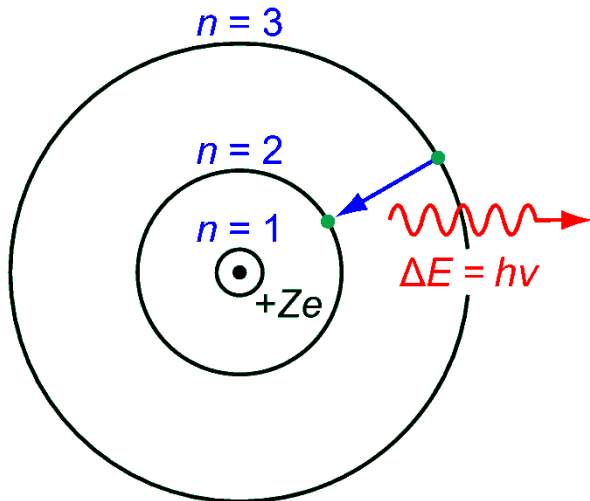
The oscillation with $n=1$ is called the **fundamental or first harmonic** (基谐波、一次谐波), and higher values of n correspond to **higher harmonics**

De Broglie Waves



De Broglie realized: **standing waves are examples of quantization.** (only certain discrete vibrational modes, characterized by the “quantum number” n , are allowed for the vibrating string.)

$$n \frac{\lambda}{2} = L \quad n = 1, 2, 3, \dots$$



De Broglie suggested: the quantization of energy in a one-electron atom have the same origin. (**the electron might be associated with a circular standing wave oscillating about the nucleus of the atom.**)

$$n\lambda = 2\pi r \quad n = 1, 2, 3, \dots$$



Louis de Broglie
(1892–1987)

The wavelength of electron

(1) Bohr 1913: angular momentum

$$L = m_e v r = n \frac{h}{2\pi}$$

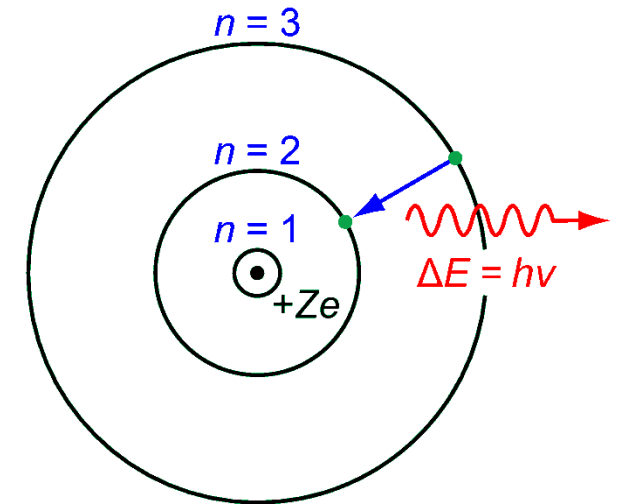
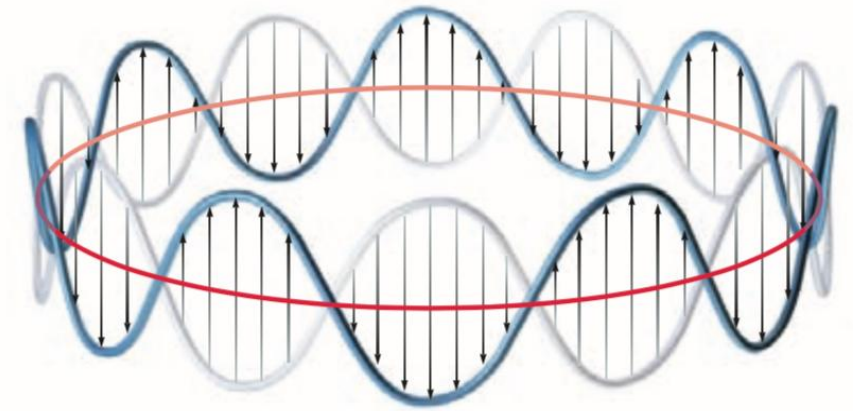
(2) de Broglie 1924:

$$2\pi r = n\lambda$$



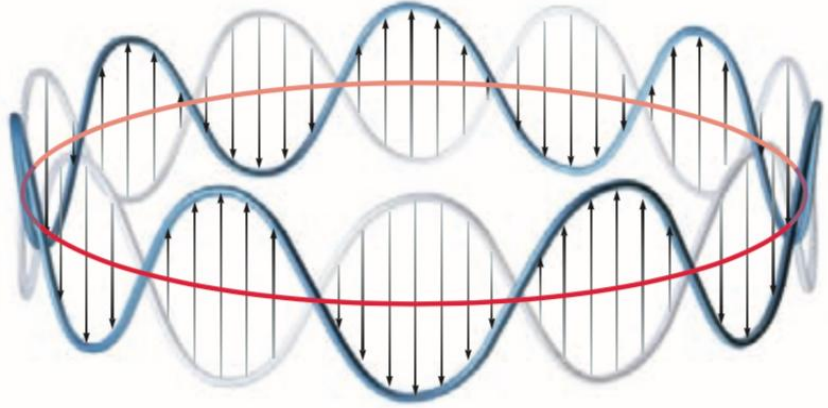
$$\lambda = \frac{h}{m_e v} = \frac{h}{p}$$

relationship holds for **both electron and photon**



De Broglie proposed that any particle moving with magnitude of linear momentum p has wavelike properties and a wavelength of h/p associated with its motion.

Indeterminacy and Uncertainty: The Heisenberg Principle



The angular position is **indeterminate** because it has no definite value: the wave is spread uniformly around the circle, we cannot specify the angular position of the electron on the circle at all.

The Heisenberg indeterminacy principle: when we measure two properties, A and B, the product of which has dimensions of action, we will obtain a spread in results of each identified by ΔA and ΔB that will satisfy the following condition:

$$(\Delta A)(\Delta B) \geq h/4\pi$$

Length (position) $\times$ momentum
energy $\times$ time



Werner Heisenberg
(1901-1976)

The Schrödinger equation

Find **a wave equation** that **describes the properties of matter on the atomic scale**

In 1926, Schrödinger proposed an equation to for finding the wave function of any system: **the Schrödinger equation**.

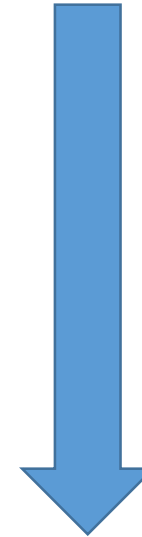
$$\hat{H}\Psi = i\hbar \frac{\partial}{\partial t} \Psi$$

- Hamiltonian (哈密顿算符)

$$\hat{H} = \hat{T} + \hat{V} = \frac{\hat{p}^2}{2m} + V = -\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}, t).$$

- Reduced Planck's constant (h-cross or h-bar 约化普朗克常数) $\hbar = h/(2\pi)$

$$\sum_{i=1}^n \frac{\partial^2}{\partial x_i^2}$$



Solve wave function: ψ

The Schrödinger equation

the Schrödinger equation

$$\hat{H}\Psi = i\hbar\frac{\partial}{\partial t}\Psi$$

the Schrödinger equation (time-independent)

$$\hat{H}\psi = E\psi$$

$$\hat{H} = \hat{T} + \hat{V} = \frac{\hat{p}^2}{2m} + V = -\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r}, t).$$



$$\sum_{i=1}^n \frac{\partial^2}{\partial x_i^2}$$

the Schrödinger equation (time-independent, 1D)

$$-\frac{h^2}{8\pi^2m} \frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

The Schrödinger equation

$$-\frac{h^2}{8\pi^2m} \frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

Example: for a particle (classical momentum= p) is associated with a wave of wavelength $\lambda=h/p$.

The wave functions could be $\psi(x) = A \sin \frac{2\pi x}{\lambda}$ and $\psi(x) = B \cos \frac{2\pi x}{\lambda}$

first derivative

$$\frac{d\psi(x)}{dx} = A \frac{2\pi}{\lambda} \cos \frac{2\pi x}{\lambda}$$

second derivative

$$\frac{d^2\psi(x)}{dx^2} = -A \left(\frac{2\pi}{\lambda} \right)^2 \sin \frac{2\pi x}{\lambda}$$

$$\frac{d^2\psi(x)}{dx^2} = - \left(\frac{2\pi}{\lambda} \right)^2 \psi(x)$$

de Broglie relation

$$\frac{d^2\psi(x)}{dx^2} = - \left(\frac{2\pi}{h} p \right)^2 \psi(x) \Rightarrow -\frac{h^2}{8\pi^2m} \frac{d^2\psi(x)}{dx^2} = \frac{p^2}{2m} \psi(x)$$

The Schrödinger equation

$$-\frac{h^2}{8\pi^2m} \frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

$$-\frac{h^2}{8\pi^2m} \frac{d^2\psi(x)}{dx^2} = \frac{p^2}{2m} \psi(x)$$

kinetic energy $\mathcal{T} = p^2/2m$



$$-\frac{h^2}{8\pi^2m} \frac{d^2\psi(x)}{dx^2} = \frac{p^2}{2m} \psi(x) = \mathcal{T}\psi(x)$$

$E = \mathcal{T} + V(x)$ kinetic energy + potential energy



$$-\frac{h^2}{8\pi^2m} \frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

time-independent Schrödinger equation for a particle moving in one dimension.

$$\psi(x) = A \sin \frac{2\pi x}{\lambda} \quad \text{and} \quad \psi(x) = B \cos \frac{2\pi x}{\lambda}$$



$$-\frac{h^2}{8\pi^2m} \frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

Interpretation of the wave function in the Schrödinger equation

What is the **physical meaning of the wave function ψ** ? (no way of measuring ψ directly)

classical theory of electromagnetism (wave view):

- No direct way to measure the amplitudes of the electric and magnetic fields of light wave.
- Intensity of the light wave (can be measured) is proportional to the square of the amplitude of the electric field: $\text{intensity} \propto (E_{\text{max}})^2$

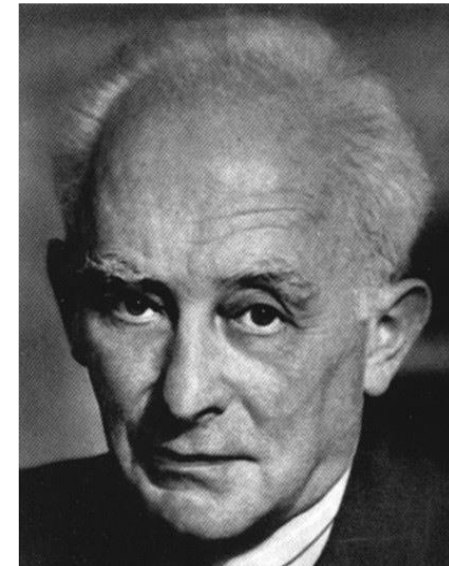
Quantum mechanics (particle view):

- the intensity is proportional to the density of photons in a region of space.

Connecting the **wave and particle views of the electromagnetic field**: probability of finding a photon is given by the square of the amplitude of the electric field

Probability of finding a particle is given by the square of the amplitude of the wave function

Probability: $P(x) = |\psi(x)|^2$



Max Born
(1882-1970)

Interpretation of the wave function in the Schrödinger equation

The **probabilistic interpretation requires** that any function must meet **three mathematical conditions** before it can be used as a wave function.

“单值”

- The probability density must be **normalized**, ensures that probability density is properly defined, and that all possible outcomes are included (平方可积)

$$\int_{-\infty}^{+\infty} P(x) dx = \int_{-\infty}^{+\infty} [\psi(x)]^2 dx = 1$$

- $P(x)$ must be **continuous** at each point x . (连续)

At some specific point, its value must be the same regardless approached from the left or from the right. This translates into the condition that $\psi(x)$ and its first derivative $\psi'(x)$ are continuous.

- $P(x)$ must be **bounded** at large values of x . (有限)

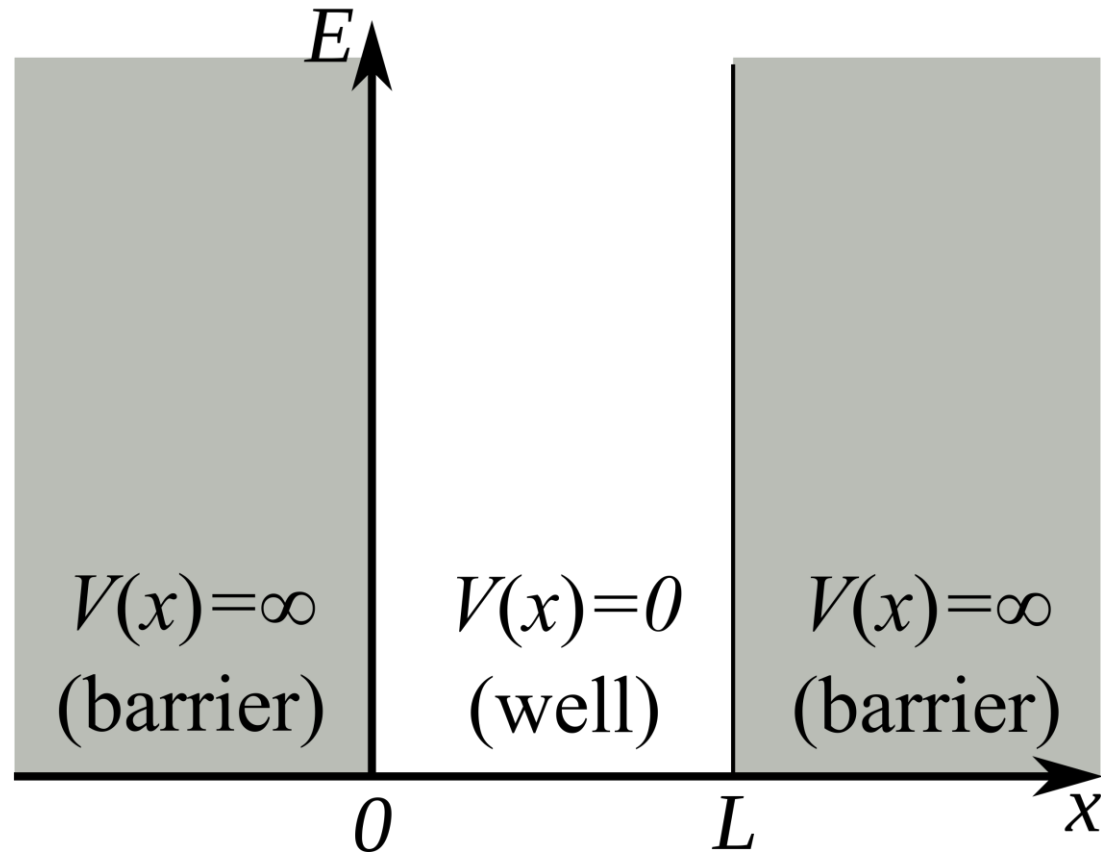
$$\psi \longrightarrow 0 \quad \text{as} \quad x \longrightarrow \pm \infty$$

Particle-in-a-Box models

$$-\frac{h^2}{8\pi^2m} \frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

Particle-in-a-Box models: a particle confined by potential energy barriers to a certain region of space.

1D box: the particle is located on the x-axis in the interval between 0 and L (L is the length of the box)



- The potential energy $V(x)$ rise abruptly to an infinite value at the two end walls to prevent fast-moving particles from escaping.
- Inside the box, the motion of the particle is free, so $V(x) = 0$

$$V(x) = \begin{cases} 0, & x_c - \frac{L}{2} < x < x_c + \frac{L}{2}, \\ \infty, & \text{otherwise,} \end{cases},$$

$$\psi(x) = 0 \text{ for } x \leq 0 \text{ and } x \geq L$$

Particle-in-a-Box models

Schrödinger Equation for a particle in a box:

$$-\frac{h^2}{8\pi^2m} \frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

Inside the box, where $V = 0$, the Schrödinger equation has the following form:

$$-\frac{h^2}{8\pi^2m} \frac{d^2\psi(x)}{dx^2} = E\psi(x)$$

or equivalently,

$$\frac{d^2\psi(x)}{dx^2} = -\frac{8\pi^2mE}{h^2} \psi(x)$$

Particle-in-a-Box models

a potential solution?

$$\psi(x) = A \sin kx$$

- The boundary conditions require: $\psi(x) = 0$ at $x = 0$ 
 - The cosine function can be eliminated, $\cos(0) = 1$
 - The sine function meets the requirement, $\sin(0) = 0$

- The boundary conditions require that $\psi(x) = 0$ at $x = L$  $\psi(L) = 0$
$$\psi(L) = A \sin kL = 0$$
$$kL = n\pi \quad n = 1, 2, 3, \dots$$

$$\psi(x) = A \sin \left(\frac{n\pi x}{L} \right) \quad n = 1, 2, 3, \dots$$

Particle-in-a-Box models

The wave function must be normalized

$$A^2 \int_0^L \sin^2 \left(\frac{n\pi x}{L} \right) dx = 1$$



$$A^2 \left(\frac{L}{2} \right) = 1 \quad A = \sqrt{\frac{2}{L}}$$

The normalized wave function for the one-dimensional particle in a box is

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin \left(\frac{n\pi x}{L} \right) \quad n = 1, 2, 3, \dots$$

Particle-in-a-Box models

$$-\frac{h^2}{8\pi^2m} \frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

$$\frac{d^2\psi(x)}{dx^2} = -\frac{8\pi^2mE}{h^2} \psi(x)$$



$$\frac{8\pi^2mE_n}{h^2} = \frac{n^2\pi^2}{L^2}$$

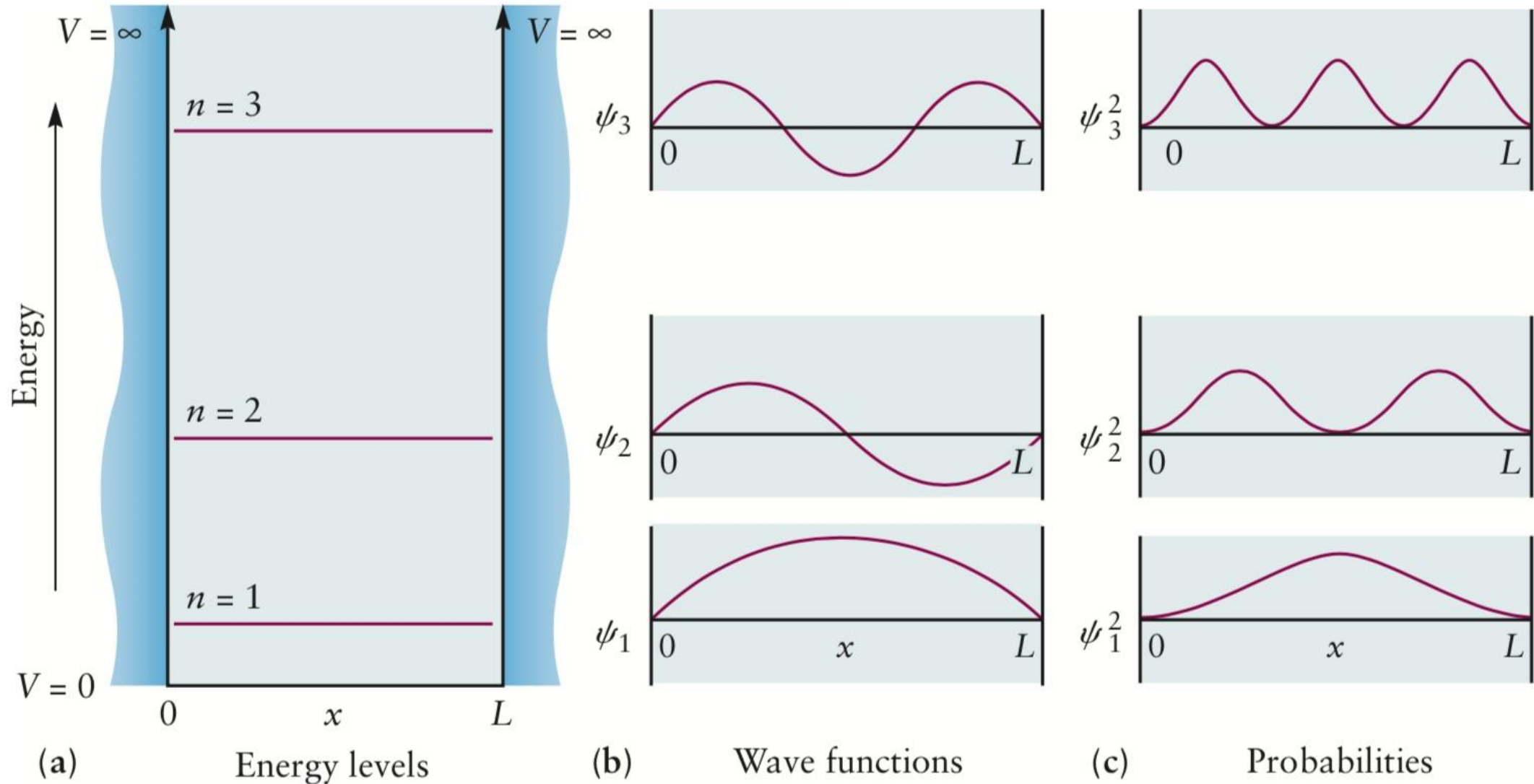
$$E_n = \frac{n^2h^2}{8mL^2} \quad n = 1, 2, 3, \dots$$

$$\begin{aligned} \frac{d^2\psi_n(x)}{dx^2} &= \frac{d^2}{dx^2} \left[\sqrt{\frac{2}{L}} \sin \left(\frac{n\pi x}{L} \right) \right] \\ &= -\left(\frac{n\pi}{L} \right)^2 \left[\sqrt{\frac{2}{L}} \sin \left(\frac{n\pi x}{L} \right) \right] \\ &= -\left(\frac{n\pi}{L} \right)^2 \psi_n(x) \end{aligned}$$

The **energy** of a particle in the box **is quantized**

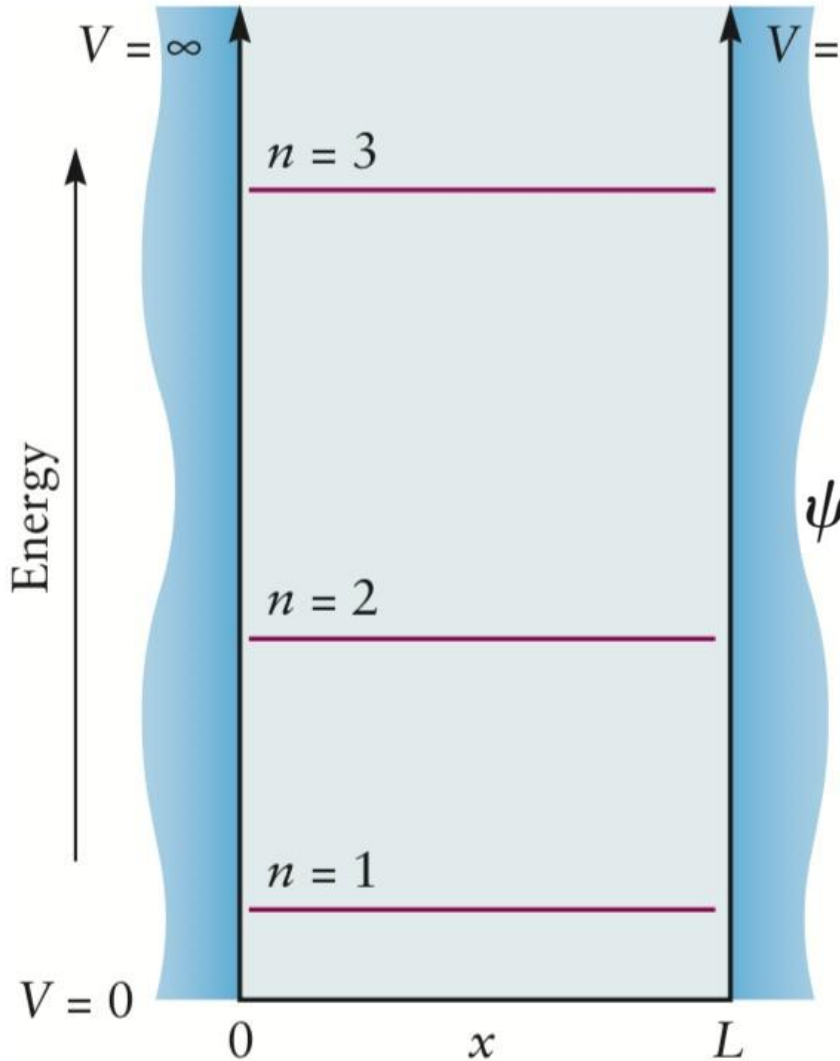
Particle-in-a-Box models

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \quad n = 1, 2, 3, \dots$$



The wave function ψ_n has $n - 1$ nodes

Particle-in-a-Box models



Can a particle in a box have zero energy?

If $n = 0$; $E_0=0$;

$\psi_n(x) = A \sin (n\pi x/L)$ makes $\psi_0(x)$ zero everywhere

no particle in the box

(a)

Energy levels

Particle-in-a-Box models

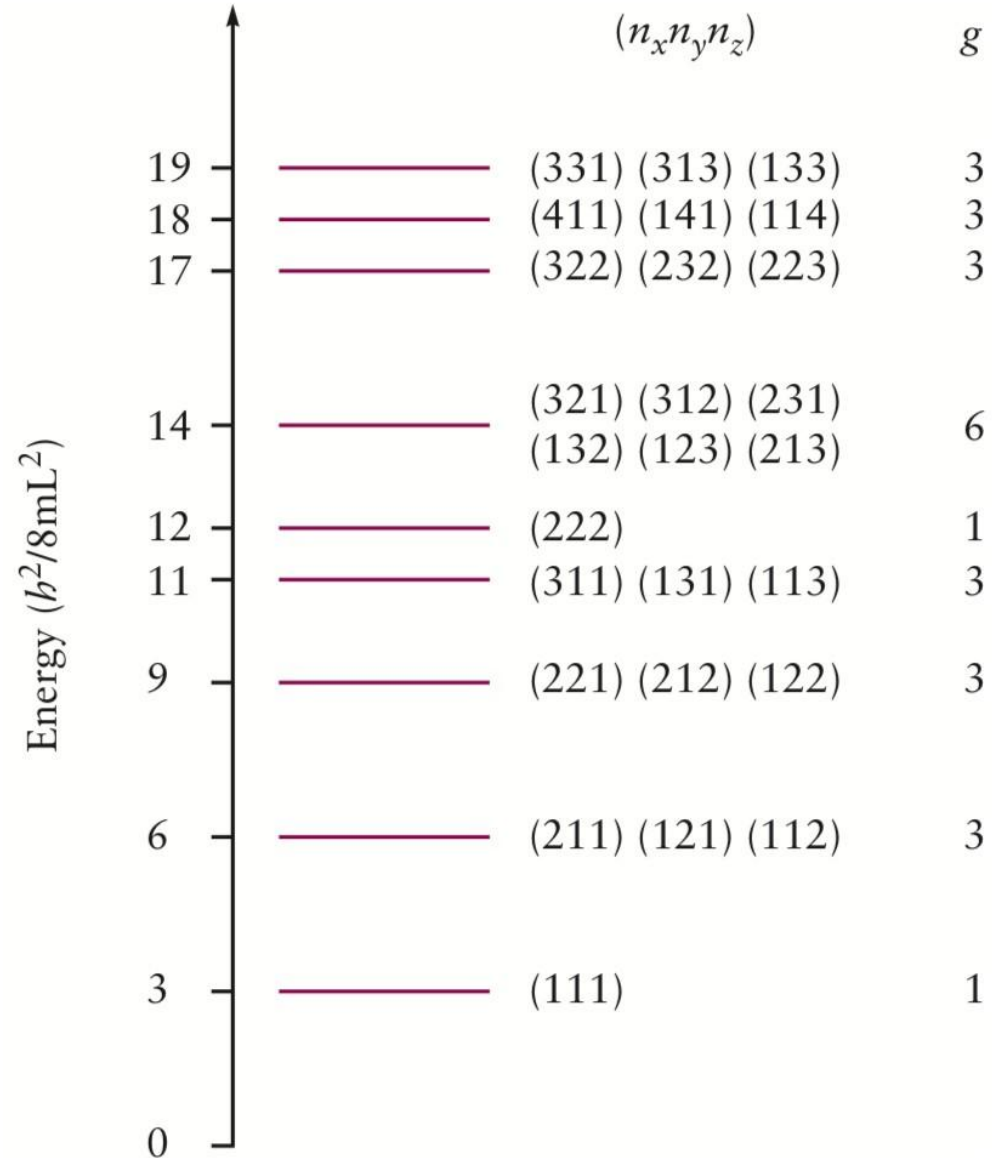
3-D boxes?

$$E_{n_x n_y n_z} = \frac{h^2}{8m} \left[\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2} \right] \quad \begin{cases} n_x = 1, 2, 3, \dots \\ n_y = 1, 2, 3, \dots \\ n_z = 1, 2, 3, \dots \end{cases}$$

Cubic box by setting $L_x = L_y = L_z = L$.

$$E_{n_x n_y n_z} = \frac{h^2}{8mL^2} [n_x^2 + n_y^2 + n_z^2] \quad \begin{cases} n_x = 1, 2, 3, \dots \\ n_y = 1, 2, 3, \dots \\ n_z = 1, 2, 3, \dots \end{cases}$$

Particle-in-a-Box models



- Certain energy values appear more than once (squares of different sets of quantum numbers can add up to give the same total).
- Such energy levels, which correspond to more than one quantum state, are called **degenerate** (简并) .

Major points of today's lecture

- **Introduction to quantum mechanics**

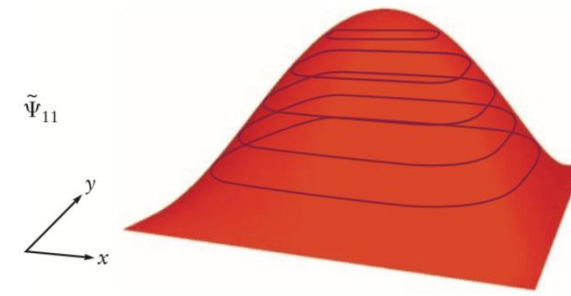
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7. Wave functions for particles in two- and three-dimensional boxes

Particle in square boxes

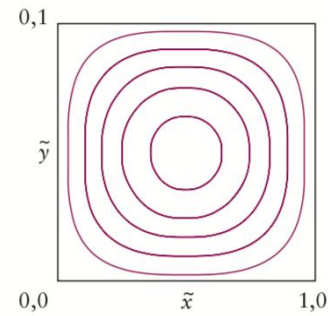
$$\Psi_{n_x n_y}(x, y) = \psi_{n_x}(x)\psi_{n_y}(y) = \frac{2}{L} \sin\left(\frac{n_x \pi x}{L}\right) \sin\left(\frac{n_y \pi y}{L}\right)$$

The normalized wave function for the one-dimensional particle in a box is

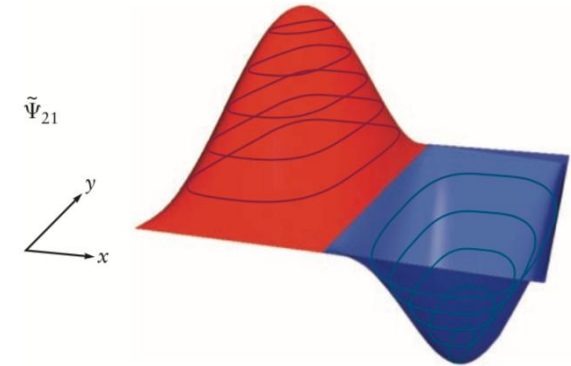
$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \quad n = 1, 2, 3, \dots$$



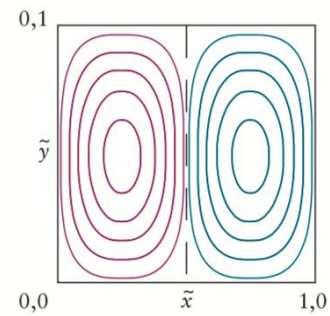
(a)



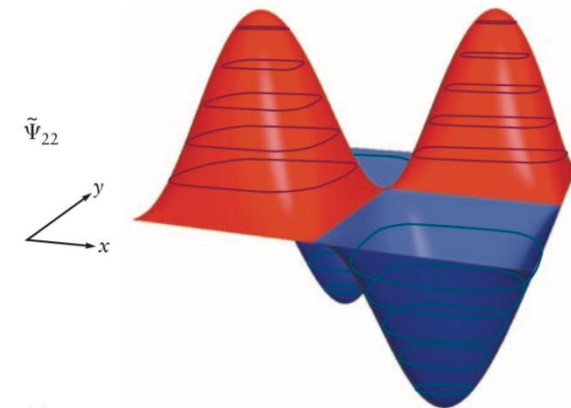
(b)



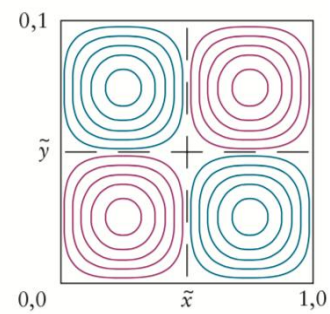
(c)



(d)



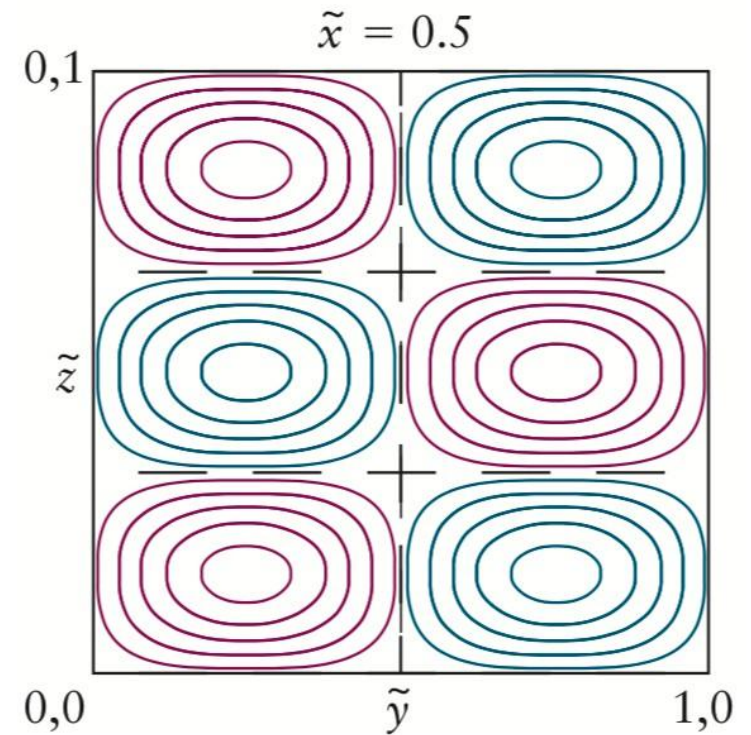
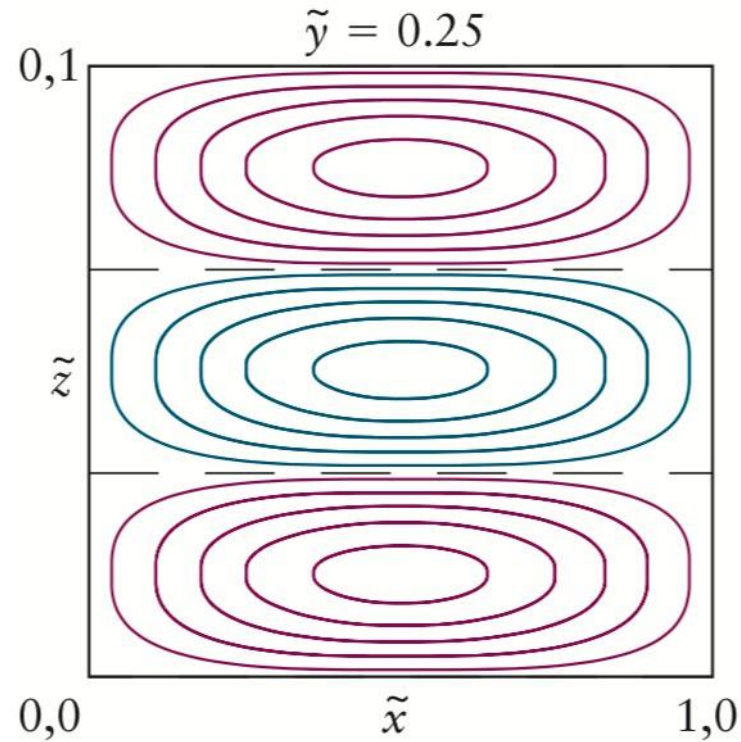
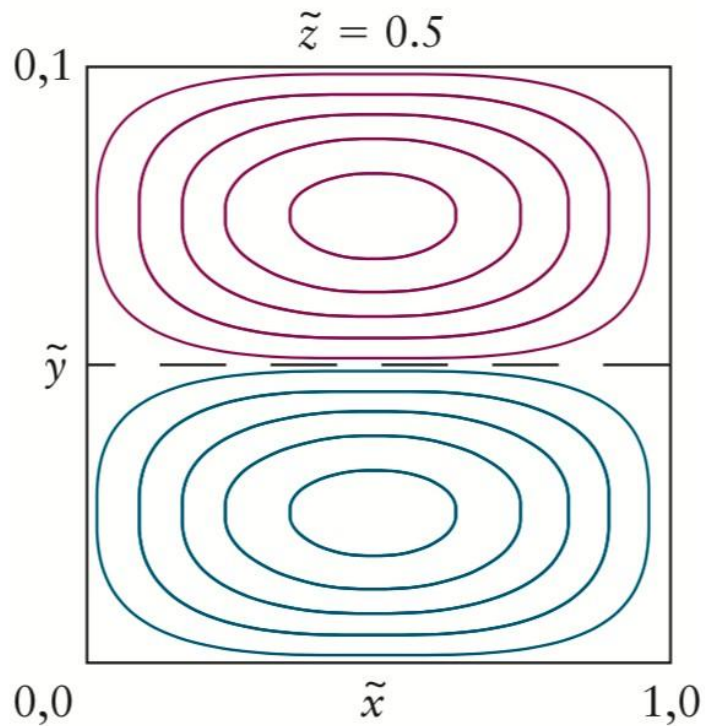
(e)



(f)

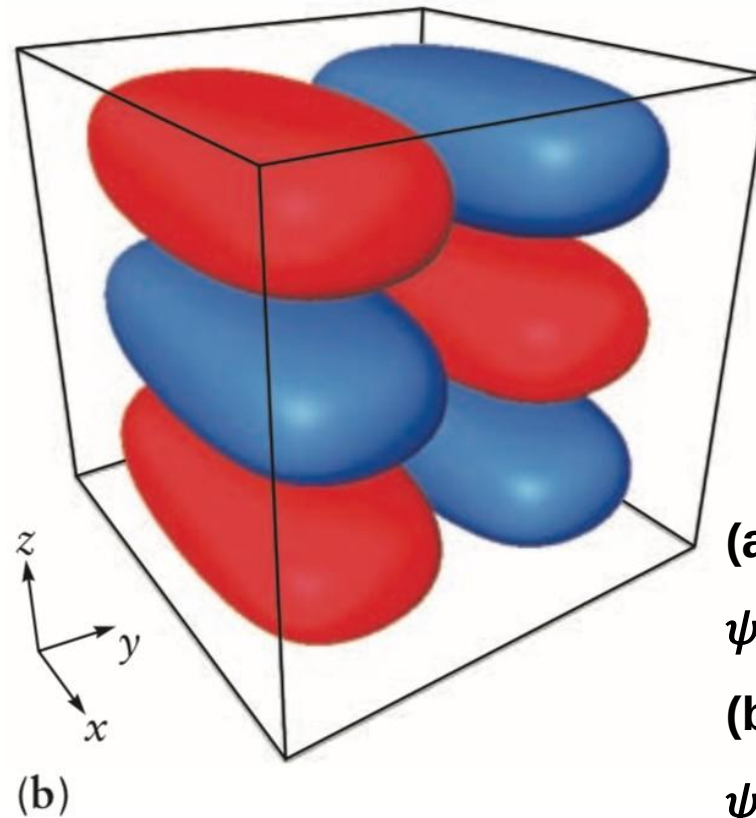
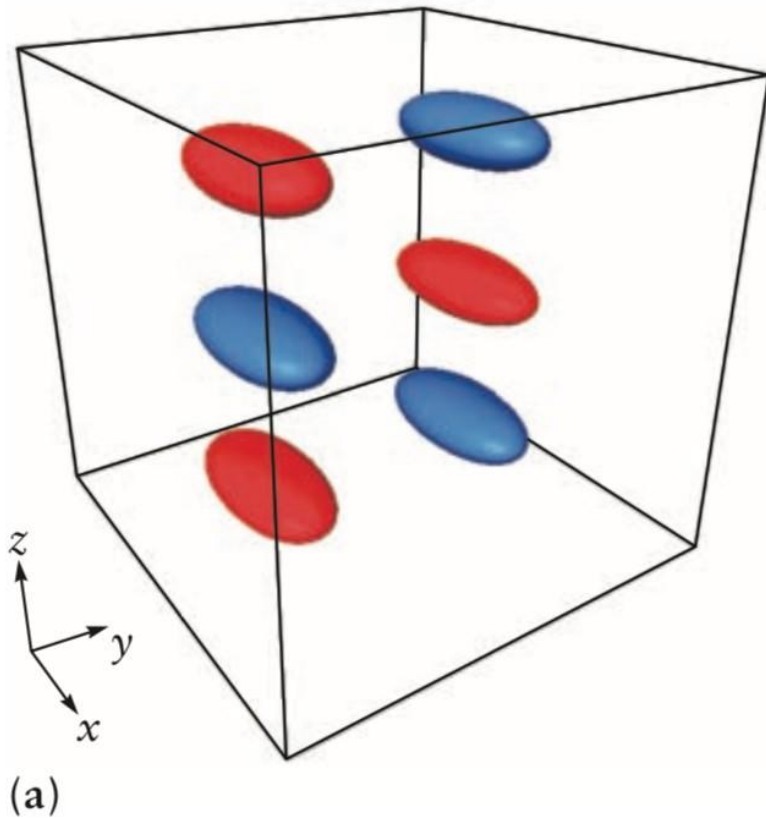
Particle in cubic boxes

$$\Psi_{n_x n_y n_z}(x, y, z) = \left(\frac{2}{L}\right)^{3/2} \sin\left(\frac{n_x \pi x}{L}\right) \sin\left(\frac{n_y \pi y}{L}\right) \sin\left(\frac{n_z \pi z}{L}\right)$$



Particle in cubic boxes

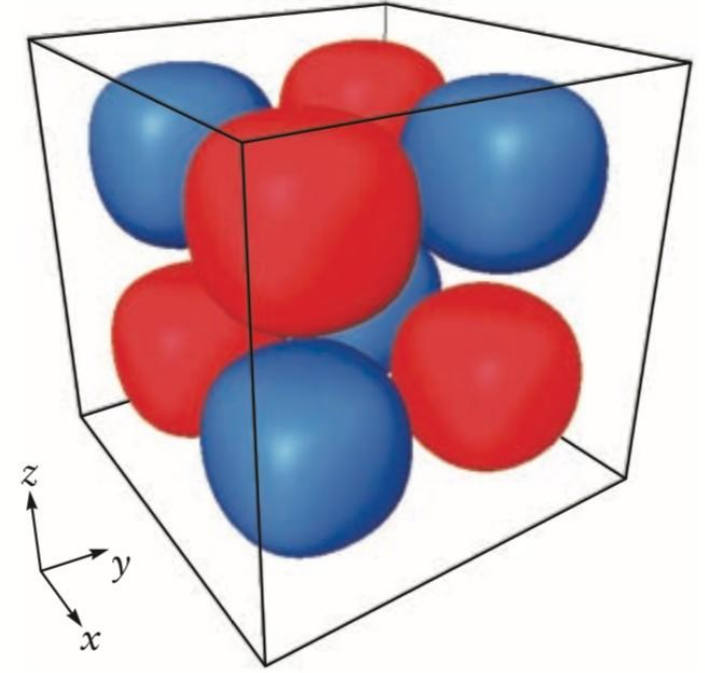
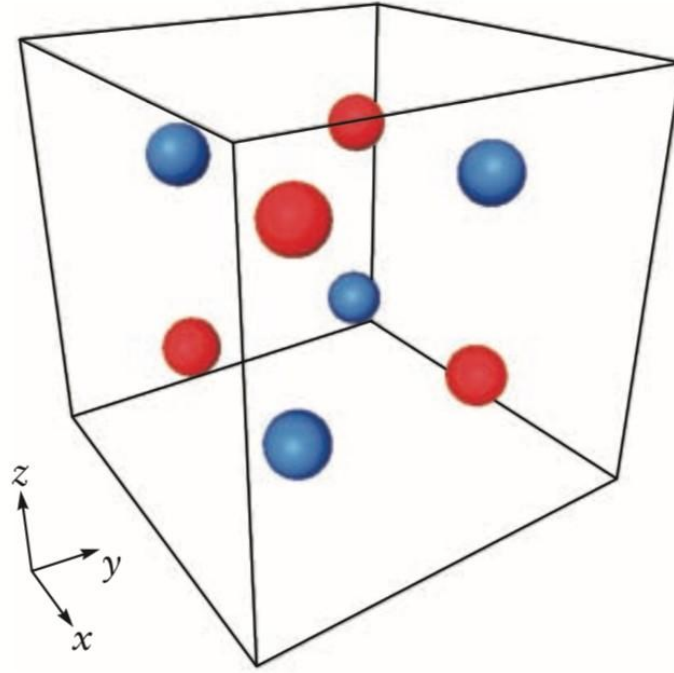
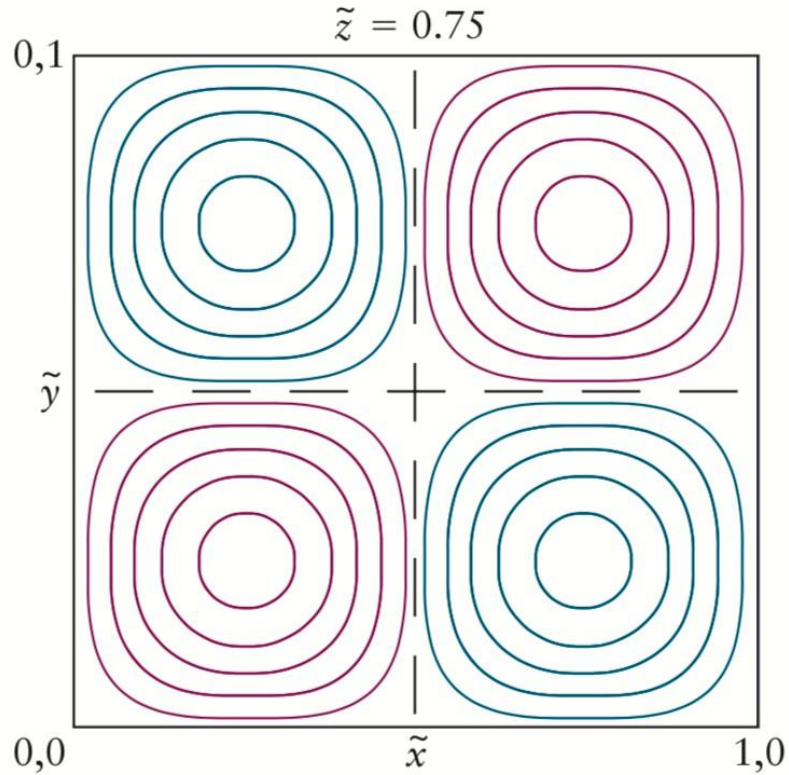
Isosurfaces (等值面): the wave function has constant value at each point on them.



(a) Isosurfaces for wave function value $\psi_{123} = \pm 0.8$.

(b) Isosurfaces for wave function value $\psi_{123} = \pm 0.2$.

Particle in cubic boxes



Representations of $\psi_{222}(x, y, z)$ for a particle in a cubic box.

(a) Contour plots for a cut taken at $z = 0.75$.

(b) Isosurfaces for wave function value $\psi_{222} = \pm 0.9$.

(c) Isosurfaces for wave function value $\psi_{222} = \pm 0.3$.

Homework: read “principle of modern chemistry”, chapter 4

“in search of Schrödinger's cat” by John Gribbin



“I don’t like it, and I’m sorry I ever had anything to do with it.”

Erwin Schrödinger